MILITARY SPECIFICATION

TELESCOPE, ELBOW: 11733780

1. SCOPE

1.1 This specification covers one type of a one and one half power (1.5X} fixed focus telescope known as Telescope Elbow: 11733780 used with the Lightweight Company Mortar, 60MM, XM224 (See 6.1). Throughtout this specification this instrument is referenced as "telexcope". This telescope has a radioactively illuminated reticle.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-L-50778	Liquid Scintillation System for Radioactive Fire Control

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila., Pa. 19137) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this documentor tor by letter.

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MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods
MIL-STD-1458	Radioactive Materials: Marking and Labeling of Items, Packages and Shipping Containers for Identification in Use Storage and Transportation

DRAWING

U.S. Army, Frankford Arsenal

F5549108	Telescope, Collimating
F8565556	Pressure Tester Two Station
D7680617	Shock Tester
F11733780	Telescope, Elbow, 11733780

PACKAGING DATA SHEETS

11733780	Packaging of Telescope, Elbow, 11733780

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 CFR 171-190- Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory)
Part 20: Standards for Protection against radiation
Part 30: Rules of General applicability to Licensing of by-product material.

(Applicaiton for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

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3. REQUIREMENTS

3.1 Radioactive material. - Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, possession, handling and shipping of radioactive material.

3.1.1 Regulations pertaining to radioactive devices.- Marking, identification, handling, and shipping of radioactive devices shall be in accordance with MIL-STD-1458.

3.1.2 External tritium Contamination. - Any part of tritium shallnot be present on the external surfaces of the telescope in excess of 1000 dpm/ 100 sq. cm. subsequant to each of these requirements: 3.7.1, 3.7.2, 3.7.3, and 3.7.4.

3.1.3 Illumination check. - Illuminated areas shall be given a visual check to determine if the litght source is illuminating subsequent to requirements 3.7.1, 3.7.2, 3.7.3, and 3.7.4.

3.2 Materials. - Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specificaiton.

3.3 Fabrication. - The telescope shall be manufactured in accordance wiht Drawing F11733780.

3.4 General specifications. - The following requirements of MIL-F 13926 shall be met:

a. Order of precedence.
b. Dimensions and tolerances (see USASI Y14.5).
c. Inorganic protective surface finishes.
d. Parts identification and marking.
e. Workmanship.
f. Bonded joints.

3.5 Temperature conditions. - The requirements of this specification shall be met with the telescope thermally stabilized at ambient temperature of between 60° and 90° Fahrenheit (F) unless otherwise specified in the requirement paragraphs.

3.6 Workmanship. - The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surfface finishes shall be free of any defects shich effect their protective qualitites or appearances. surfaces shall be clean and free of any scratches or blemishes which may effect their optical qualities and shall meet the requirements of 3.7.5 and 3.7.6.

3.7 Environment. - Any telescope subjected to any or all of the requirements of3.7.1, 3.7.2 and 3.7.3, inclusive shall subsequently meet the requirements of 3.7.4 to 3.8.6 inclusive, unless otherwise specified.

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3.7.1 Storage temperature. - The telescope shall show no evidence of physical failure such as cement separation, glass breakage or casting failure after having been exposed and thermally stabilized at ambient temperature of -50° and +160°F (see 3.7.4.1).

3.7.2 Shock. - The telescope shall be positioned with the line of sight through eyepiece and objective tube in a horizontal plane and the telescope supported at the 0.624 inch diameter of the objective tube. The telescope shall be subjected to six shocks each at 200g for one (1) mini-second. The shock pulse shall be a 1/2 sine wave.

3.7.3 Vibration. - The telescope shall be positioned with the line of sight through the eyepiece and objective tube in a horizontal plane and the telescope supported at the 0.624 inch diameter of the objective tube. The telescope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds.

3.7.4 Sealing and flushing. - The telescope shall show no evidence of leakage in-excess of 0.2 psig when subjected to en internal pressure of 5 ± 0.10 psig for a period of two hours using dry nitrogen having a dew point at least as low as -25°F. The telescope shall then be flushed with dry nitrogen until the interior atmosphere of the telescope has a dew point at least as low as -25°F.

3.7.4.1 Condensation (low temperature). - When viewed through the eyepiece and objective ends of the telescope, there shall be no evidence of condensation at room temperature (60° to 90°F) immediately upon thermal stabilization, subsequent to being subjected to the -50°F and prior to the +160°F requirement of 3.7.1.

3.7.5 Cleanliness. - When viewed through the eyepiece, the surface of the reticle shall show no more than six particles of dirt. The maximum width of each particle shall not exceed the apparent width of the reticle line. The separation-between each dirt particle or foreign matter shall be at least 10 mils.

3.7.6 Defects. - There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, or adhesive separations on any glass component when the telescope is viewed through both the eyepiece and objective" ends.

3.8 Accuracy.

3.8.1 Collimation. - The optical axis of the telescope shall be in coincidence with the geometric axis within 1.0 mil when the telescope is rotated 3200 mils (90° to the left and 90° to the right from a vertical position of the eyepiece) about the geometric axis as defined by the exterior mounting surface of the objective tube of the telescope.

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3.8.2 Parallax. - Parallax at the optical axis shall not exceed 0.5 mil when viewing a target between 40 meters and infinity.

3.8.3 Resolution. - The resolution on the optical axis shall be 30 seconds of arc or less at one focus setting of an auxiliary observation telescope positioned at the eyepiece.

3.8.4 Parallelism of reticle and image. The verticle line shall be parallel to the image of a plumb line within 1 degree of arc when each of the three indices (90° apart) are in position for coincidence with a vertical plane index through the geometric axis of the telescope's objective tube.

3.8.5 Eyepiece focus. - The eyepiece focus of the reticle shall be set between minus .50 and minus 1.25 diopter.

3.8.6 Illumination. - The reticle lines shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into night.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) telescopes plus three (3) each of all items covered by SQAP. The three telescopes shall be tested as specified herein for all the requirements of this specification. The items covered by SQAP shall be tested, as specified thereon, for all characteristics delineated on the SQAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of-this inspection with the sample, including certificates of conformance for materials.

4.2.2 contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and Subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP'S, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Telescope. - Examination and tests related to Section 3 herein shall be performed on a class basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLAss CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDEDURE
101. Vibration	3.7.3	4.6.3
102. Sealing and flushing	3.7.4	4.6.4
103. External tritium contamination	3.1.2	4.6.11
104. Illumination check	3.1.3	Visual
105. Workmanship	3.6	4.6.13
106. Cleanliness	3.7.5	4.6.6
107. Defects	3.7.6	4.6.7

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108.	Collimation	3.8.1	4.6.8.1
109.	Parallax	3.8.2	4.6.8.2
110.	Resolution	3.8.3	4.6.9
111.	Parallelism of reticle and image	3.8.4	4.6.8.3
112.	Eyepiece- focus	3.8.5	4.6.10
113.	Illumination	3.8.6	4.6.12

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between +60°F and +90°F.

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. - One telescope shall be selected at random as a special sampl from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawing - visual
302.	General specification	3.2	MIL-F-13926- visual

NOTE: The tests in Table II shall be conducted at a temperature between +60° and +90°F.

4.4.2 Environmental. - Three telescopes shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III, and shall then meet the requirements and tests in Table I.

TABLE III - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-50°F and +160°F)	3.7.1	4.6.1
304.	condensation, post low temperature cycle	3.7.4.1	4.6.5
305.	Shock	3.7.2	4.6.2
306.	External tritium contamination (Post temperature cycle, Post shock)	3.1.2	4.6.11
307.	Illumination check (Post (temperature cycle, Post shock)	3.1.3.	Visual

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NOTE: The test of Characteristics 304 through 307 inclusive shall be conducted. at a temperature between +60°F and +90°F.

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved. (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - All inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.1 Contractor design. - In the absence of Government inspection equipment design the contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in-design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing. - The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F. The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall not be so located that radient heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to

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determine or control the specified chamber temperature shall be centrally located within the test chamber in the SUPPIY airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.5.2.3.2 Accuracy. - Special testing equipment to test the accuracy of the telescope shall consist of a telescope holding device and a target projector collimator. The test equipment shall conform to the following requirements:

a. Telescope holding device. - This device shall be capable of securing and maintaining the exterior mounting surface of the telescope's objective tube horizontal within 20 seconds of arc. The device shall provide sufficient tension to the objective tube to permit it to be maintained in a "zero" rotational position while the telescopes mechanical axis lays in a vertical plane. This holding device shall also permit the telescope to be rotated through the excursion specified in 3.8.1 and 3.8.4 without slippage, or rotation under its own weight. This holding device shall also be equipped with three telescope position indices. One index shall indicate the "zero" rotational position of the telescope within plus or minus 3 minutes, the second index shall be positioned 900 ± 0°3' clockwise from the vertical plane, the third index shall be positioned 90° ± 0°3' counterclockwise from the vertical plane.

b. Target projector collimator. - The target projector collimator shall have a clear aperture of at least 1.5 inches. The target reticle of the collimator shall contain a 1.0 mil diameter circle at the intersecting point of the horizontal and vertical reticle lines to determine conformance with collimation, as specified in 3.8.1. Tolerance marks shall also be incorporated on both sides of the vertical reticle line to determine conformance to parallelism of reticle and image, as specified in 3.8.4. Unless otherwise specified, the target collimator shall be set at infinity focus. The target collimator shall be positioned and aligned so that its optical axis is adjacent to, and parallel within 40 seconds to the geometric axis of the objective tube of the telescope under test. The vertical reticle line of the target collimator shall be adjusted plumb.

4.5.2.3.3 Sealing. - Special testing equipment to test sealing of the telescope shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester - Two Station. A special adapter shall be utilized when applying, or flushing, the nitrogen gas in the telescope.

4.6 Test methods and procedures.

4.6.1 Storage temperature. - Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.l. The telescope shall be placed in the test chamber and the temperature of the chamber reduced, over a period of 4 hours plus or minus 15 minutes, from room ambient temperature (60° to 90°F) to -50°F. The telescope shall be maintained at

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this extreme temperature for a minimum of 5 hours. upon completion of the 5 hour exposure the temperature of the chamber shall be raised, over a period of 4 hours plus or minus 15 minutes, to room temperature (+60° to 90°F). The telescope shall immediately removed from the test chamber and subjected to the test specified in 4.6.5 for evidence of moisture (condensation). The telescope shall meet the requirements of 3.7.4.1. The telescope shall again be placed in the test chamber and the temperature of the chamber raised, over a period of 4 hours ± 15 minutes, to +160°F. The telescope shall remain at this temperature for a minimum of 5 hours. Upon completion of the 5 hour exposure the temperature of the chamber shall be reduced, over a period of 4 hours plus or minus 15 minutes, to room ambient temperature. The telescope shall then be removed from the test chamber and subjected to the tests in 4.6.11, and then a visual examination for reticle illumination. It shall-meet the requirements of 3.1.2 and 3.1.3 respectively. The telescope shall be subjected to a visual, tactile-and audible examination. It shall show no evidence of physical damage. The telescope shall then be subjected to the tests in 4.6.4 through 4.6.12(except 4.6.11) to determine overall conformance to the requirements of 3.7.

4.6.2 Shock. - Conduct the shock test with the telescope positioned on an adapter affixed to a shock test machine that conforms to the requirements of Drawing C7680617. The adapter shall be capable of holding and positioning the telescope as specified in 3.7.2. The test shall be performed in accordance with Test Method 516.2, Procedure IV of MIL-STD-81O, except that one shock shall be applied to the telescope in each direction along each of three mutually perpendicular axes (total of 6 shocks). The shock pulse shape, peak value and normal time duration shall be as specified in 3.7.2. Subsequent to the 6 shocks the telescope shall be removed from the shock test adapter and subjected to the test in 4.6.11, and then a visual examination for reticle illumination. It shall mmet the requirements of 3.1.2 and 3.1.3 respectively. The telescope shall be subjected to a visual, tactile and audible examination. It shall show no evidence of physical damage. The telescope shall then be subjected to the tests in 4.6.4 through 4.6.12 (except 4.6.11) to determine overall conformance to the requirements of 3.7.

4.6.3 Vibration. - Conduct the vibration test with the telescope positioned on an appropriate adapter affixed to a standard vibrating machine that is capable of providing the amplitude and frequency specified in 3.7.3. Unless otherwise specified the test equipment shall be in accordance with the "Test Facilities” requirements of MIL-F-13926. The applied motion shall be in a vertical plane for the time period specified. Upon completion of the vibration cycle the telescope shall be removed from the vibration adapter and subjected to the test in 4.6.11, and then a visual examination for reticle illumination. It shall meet the requirements of 3.1.2 and 3.1.3 respectively. The telescope shall be subjected to a visual, tactile and audible examination. There shall be no evidence of physical damage. The telescope shall then be subjected to the tests in 4.6.4 through 4.6.12 (except 4.6.11) to determine overall conformance to the requirements of 3.7.

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4.6.4 Sealing and flushing. - Internal pressurizing of the telescope to test sealing shall be performed using dry nitrogen gas having a dew point at least as low as-25°F and special testing equipmetn conforming to 4.5.2.3.3. The telescope shall be pressurized to the to the requirements of 3.7.4 in accordance with the procedure for internal pressure testing of Drawing F8565556. Maintain th aplied pressure for the period specified. With the pressure at 5 ± 0.1 psig, the teescope shall retain this pressure for 2 hours with no indication of leakage in excess of 0.2 psig as indicated on the pressure gage of the testing equipments Upon satisfactory completion of this test, the body sealing screw shall be assembled to the telescope as depicted on Drawing F11733780 immediately, to minimize intermingling of external atmosphere upon disconnection of the pressure tester adapter. teh interior atmophere of the telescope shall conform to the requirements of 3.7.4.

4.6.5 Condensation (low temperature). - Subsequent to low temperature exposure and immediately upon stabilization at room temperature (+60° to +90°F), the telescope's optical system shall be examined as specified in 3.7.4.1 for evidence of condensation (moisture). This visual examination shall employ the technique of shadowing. The telescope shall conform to the requirements of 3.7.4.1.

4.6.6 Cleanliness. - The optical surfaces of the telescope shall be inspected for cleanliness in accordance with MIL-O-13830. Inspection for dirt particles shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 foot candles. Cleanliness of the telescope shall conform to the requirements of 3.7.5.

4.6.7 Defects. - Conduct this test in sequence to the test in 4.6.6. Inspection for condensates, grease, fingerprints, fractures, adhesive separations, and etc. shall be made with the unaided eye utilizing the shadowing technique while viewing through the objective and eyepiece ends of the telescope. the optical system of the telescope shall confor to the requirements of 3.7.6.

4.6.8 Accuracy. - Testing equipment utilized in the following tests shall-conform to the requirements of 4.5.2.3.2.

4.6.8.1 Collimation. - Position the telescope on the holding device with the eyepiece in a vertical position. Observe through the eyepiece of the telescope. The intersecting point of the telescope reticle centerlines shall be in coincidence with respect to the intersecting point of the target reticle lines within the tolerance specified in 3.8.1 as determined by the 1.0 mil diameter circle of the target reticle. Repeat the collimation test procedure above when the telescope is rotated 90° to the left and 90° to the right of the original test position. The intersecting point of the telescope reticle shall be within the 1.0 mil diameter circle of the target reticle at both positions to insure overall compliance with 3.8.1.

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4.6.8.2 Parallax. - Perform this test with -the target collimator adjusted for 40 meters, then adjusted for infinity. Direct a line of sight through the telescope to the collimator target. Parallax is any apparent movement of the telescope reticle lines in relation to the target image when the observer's head is moved from side to side or up and down. Parallax shall be measured on the optical axis of the telescope (center of the reticle pattern). Displacement between the telescope reticle and the target image shall not exceed the tolerance specified in 3.8.2.

4.6.8.3 Parallelism of reticle and image. - Perform this test in conjunction with the test in 4.6.8.l. position the telescope in the vertical position ("zero" rotational position). Adjust the telescope holding device in azimuth, if necessary, to position one extremity of the telescope's verticle reticle line coincident to the target vertical line image. The reticle line shall be parallel to the target vertical line image within the tolerance specified in 3.8.4, as measured at the other extremity of the telescope's reticle line with respect to the target image tolerance marks. Repeat the above test with the telescope rotated, in its holding device, to its 90° clockwise position, then with the telescope rotated to its 90° counterclockwise position. The telescope shall meet the requirements of 3.8.4 in each of its 3 indexed positions.

4.6.9 Resolution. - Position the telescope on a holding device capable of accepting the exterior mounting surface of the objective tube of the telescope. Perform this test utilizing an auxiliary telescope with a magnification of at least 3 power (3X) and a resolving power chart on a wall target. The test methods and procedures shall be in compliance with the "Resolution Test" specified in MIL-O-13830 using a resolving power chart that contains 4 line sets as shown in Figure 1 therein. The focus of the auxiliary telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count of the resolution target. Resolution of the telescope optical system shall meet the requirements specified in 3.8.3.

4.6.10 Eyepiece focus. - A calibrated dioptometer with a magnification of at least 3 power (3X) and a diopter measuring range of zero to minus three diopter shall be utilized for measusing the eyepiece focus of the telescope reticle. Set the dioptometer objective diopter scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at the eyepiece of the telescope. Observe through the dioptometer, obtain best focus of the telescope reticle by adjustment of the dioptometer objective diopter scale. The telescope reticle shall be in best focus within the tolerance specified in 3.8.5, as measured on the dioptometer objective diopter scale.

4.6.11 External tritium contamination. - Contamination test shall be performed by wiping the telscope with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper shall be placed in a container with the proper portion of scintillation liquid. The scintillating system used to measure contamination must be calibrated and meet the minimum requirement of MIL-L-50778.

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The read out shall be within 10% of the known standard value when counted to a total of 2000 disintergrations. The actual test for determining contamination is performed by placing the container into the scintillation system-the measured conttamination shall be within the limits specified in 3.1.2.

4.6.12 Illumination. - The reticle illumination test shall be performed under simulated conditions specified in 3.8.6. Place the telescope in a dark room having a controlled light source, The observer and telescope must become dark adapted. Prior toe performing the test, place a white matted surface approximately 2 feet from the telescope window. Per for the test by observing reticle pattern and determine that this pattern is clearly discernible by means of the self-illuminating source in the dark room. Gradually raise the level of illumination to where the pattern becomes indiscernible. Center an aperture stop over the objective entrance aperture of the telescope and observe that the reticle pattern is clearly discernible. The aperture of the stop shall approximate one half (1/2) the interior diameter of the retainer for the objective lens. Raise the ambient level of illumination to 10 foot candles, remove the fixed stop and observe the reticle pattern. The reticle illumination of the telescope shall meet the requirements of 3.8.6.

4.6.13 Workmanship. - The telescope shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The telescope shall conform to the requirements of 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking of the Telescope, Elbow, 11733780 shall be in accordance with Packaging Data sheet 11733780. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Telescope, Elbow, 11733780 is used with the Mount Telescope, 11741796. These two items compose the Sight Unit XM64E2 which is used with the Lightweight Company Mortar, 60MM, XM224.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable stock numbers.
d. Applicable packaging data sheet. (See 5.1)

Custodian

Army - MU

Preparing activity

Army - MU

Project No. 1240-A700

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-48574(MU) Telescope, Elbow: 11733780

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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